

REDCap Cleaner Tutorial

Step-by-step Google Colab guide for cleaning REDCap exports into a traceable Excel workbook.

What this covers

1. Run the setup cells before using the workflow.
2. Upload a REDCap CSV, XLS, or XLSX export.
3. Configure IRB values, subject ID parsing, and column keep/rename choices.
4. Review excluded subject IDs and visit/timepoint mappings.
5. Clean, preview, and download the multi-sheet Excel workbook.

Notebook: <https://colab.research.google.com/drive/1zZG9gbXq1o3bruxNNkm54csMEjJiuBsc?usp=sharing>

 redcap_cleaner_step-by-step.ipynb  

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+ Section

Click to expand
code cells

>

Setup (ALWAYS RUN THIS FIRST!)

- Click the ▶ button below once, next to [↩ 3 cells hidden], to run all setup codes.
- Alternatively, run each of the cell: Cleaner engine, Helper functions

▶ ↩ 3 cells hidden

Click to run all code cells
in this section

>

 REDCap cleaner step-by-step

- Click the ▶ button below to execute *each step*

↩ 7 cells hidden

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REDCap cleaner step-by-step

Click the ► button below to execute *each step*

1. Upload REDCap export

Click the ► to execute this step
Either works they function the same

Choose Files No file chosen No file chosen Cancel upload

Upload the raw REDCap csv exports

REDCap cleaner step-by-step

Click the ► button below to execute *each step*

1. Upload REDCap export

[32] Show code

Choose Files 58807Sch...8_2108.csv

- 58807SchatzbergRapid-HAMD_DATA_LABELS_2026-04-28_2108.csv(text/csv) - 637048 bytes, last modified: 4/28/2026 - 100% done

Saving 58807SchatzbergRapid-HAMD_DATA_LABELS_2026-04-28_2108.csv to 58807SchatzbergRapid-HAMD_DATA_LABELS_2026-04-28_2108 (2).csv

58807SchatzbergRapid-HAMD_DATA_LABELS_2026-04-28_2108 (2).csv: 1536 rows, 71 columns.

> 1. Upload REDCap export

[12]
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Show code

Once uploaded, previews of column names and the data sheet will be displayed



58807Sch...8_2108.csv

• 58807SchatzbergRapid-HAMD_DATA_LABELS_2026-04-28_2108.csv(text/csv) - 637048 bytes, last modified: 4/28/2026 - 100% done
Saving 58807SchatzbergRapid-HAMD_DATA_LABELS_2026-04-28_2108.csv to 58807SchatzbergRapid-HAMD_DATA_LABELS_2026-04-28_2108 (3).csv
58807SchatzbergRapid-HAMD_DATA_LABELS_2026-04-28_2108 (3).csv: 1536 rows, 71 columns.

1 to 5 of 71 entries

Filter



index	shortened_column_name	original_column_name
0	col_0	Record ID
1	col_1	Event Name
2	col_2	Record/Subject ID:
3	col_3	Administered during visit number:
4	col_4	Initials of BSL staff:

Show 5 per page

1 2 10 15

Warning: Total number of columns (71) exceeds max_columns (20) limiting to first (20) columns.

891 to 900 of 1536 entries

Filter



index	col_0	col_1	col_2	col_3	col_4	col_5	col_6	col_7	col_8	col_9	col_10	col_11	col_12	col_13	col_14	col_15	col_16	col_17	col_18	col_19
890	58807_s041	Patient Contact Log (Arm 2: Baseline/Active Study/Follow-ups/Open Label)	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN
891	58807_s041	Adverse Events/Regulatory (Arm 2: Baseline/Active Study/Follow-ups/Open Label)	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN
892	58807_s041	Baseline (Visit 2) (Arm 2: Baseline/Active Study/Follow-ups/Open Label)	58807_s041	NaN	KJ	2023-11-08 14:57	NaN	Moderate	Much of the time	2.0	NaN	Moderate	Much of the time	2.0	NaN	Mild	Occasional	1.0	NaN	Mild
893	58807_s041	Study Day 1 (Visit 3) (Arm 2: Baseline/Active Study/Follow-ups/Open Label)	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN
894	58807_s041	Study Day 2 (Visit 4) (Arm 2: Baseline/Active Study/Follow-ups/Open Label)	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN
895	58807_s041	Study Day 3 (Visit 5) (Arm 2: Baseline/Active Study/Follow-ups/Open Label)	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN
896	58807_s041	Study Day 4 (Visit 6) (Arm 2: Baseline/Active Study/Follow-ups/Open Label)	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN
897	58807_s041	Study Day 5 (Visit 7) (Arm 2: Baseline/Active Study/Follow-ups/Open Label)	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN
898	58807_s041	Next Day Visit (Visit 8) (Arm 2: Baseline/Active Study/Follow-ups/Open Label)	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN
899	58807_s041	Immediate Post (Visit 9) (Arm 2: Baseline/Active Study/Follow-ups/Open Label)	58807_s041	NaN	KJ	2023-11-21 12:17	NaN	Absent	Absent	0.0	NaN	Absent	Absent	0.0	NaN	Absent	Absent	0.0	NaN	Absent

Show 10 per page

1 10 80 89 90 91 100 150 154

>

2. Customize configuration ⚙

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Show code

▼ ...

Configure cleaning

IRB(s)

58807, 54909

Enter the IRBs in this field; this information will be used to parse the subject IDs

Subject key identifier column

Record ID

You can change the column used as subject identifier by this drop down ->

▼

☒ Drop fully empty rows

Check this box if you would like to drop fully empty rows

Column keep / rename: 71 columns loaded.

Keep	Column	Alternative name
☒	Record ID	Optional new name
☒	Event Name	You can RENAME a column here
☒	Record/Subject ID:	Optional new name
☒	Administered during visit number:	Optional new name
☒	Initials of BSL staff:	Optional new name
☒	Todays date:	Optional new name
☒	1. Depression: Notes	Optional new name

> 3. Review subject ID exclusions 🤔

[20]  Show code

Run this step to preview the rows being dropped / excluded according to Step 2 configuration

▼ ... Raw subid values are read from selected column: Record ID
246 subject IDs need review. Showing first 100 below.

1 to 25 of 100 entries 

index	source_row	raw_subid	included	reason
0	2	2	false	raw identifier has fewer than 4 characters
1	3	3	false	raw identifier has fewer than 4 characters
2	4	4	false	raw identifier has fewer than 4 characters
3	5	5	false	raw identifier has fewer than 4 characters
4	6	6	false	raw identifier has fewer than 4 characters
5	7	7	false	raw identifier has fewer than 4 characters
6	8	8	false	raw identifier has fewer than 4 characters
7	9	9	false	raw identifier has fewer than 4 characters
8	10	10	false	raw identifier has fewer than 4 characters
9	11	11	false	raw identifier has fewer than 4 characters
10	12	12	false	raw identifier has fewer than 4 characters
11	13	13	false	raw identifier has fewer than 4 characters
12	14	14	false	raw identifier has fewer than 4 characters
13	15	15	false	raw identifier has fewer than 4 characters
14	16	16	false	raw identifier has fewer than 4 characters
15	17	18	false	raw identifier has fewer than 4 characters
16	18	19	false	raw identifier has fewer than 4 characters
17	19	20	false	raw identifier has fewer than 4 characters
18	20	21	false	raw identifier has fewer than 4 characters
19	21	22	false	raw identifier has fewer than 4 characters
20	22	23	false	raw identifier has fewer than 4 characters
21	1313	MDD_1001	false	cleaned identifier does not match expected s### format
22	1314	MDD_1003	false	cleaned identifier does not match expected s### format
23	1315	MDD_1003	false	cleaned identifier does not match expected s### format
24	1316	MDD_1007	false	cleaned identifier does not match expected s### format

Show per page

1 2 3 4

> 4a. Review visit/timepoint mapping 📅

[13]
✓ 1s



Show code

Run this step to verify / change the Visit labels

▼ ...

Review visit/timepoint mapping

Unique values from the selected visits column will be canonicalized in the cleaned file. Edit the final timepoint values before running step 5 if needed.

Visits column Event Name

Raw visit value

Delegation Log (Arm 3: Delegation Log)

Screening (Visit 1) (Arm 1: Screening)

Patient Contact Log (Arm 2: Baseline/Active Study/Follow-ups/Open Label)

Adverse Events/Regulatory (Arm 2: Baseline/Active Study/Follow-ups/Open Label)

Baseline (Visit 2) (Arm 2: Baseline/Active Study/Follow-ups/Open Label)

Study Day 1 (Visit 3) (Arm 2: Baseline/Active Study/Follow-ups/Open Label)

Proposed

Delegation Log (Arm 3:
Delegation Log)

V1

Patient Contact Log (Arm 2:
Baseline/Active
Study/Follow-ups/Open
Label)

Adverse Events/Regulatory
(Arm 2: Baseline/Active
Study/Follow-ups/Open
Label)

V2

V3

Use in cleaned file

If you need to change the visit name

V1

Patient Contact Log (Arm 2: Baseline/Active S

Adverse Events/Regulatory (Arm 2: Baseline/

V2

V3

>

5. Clean and preview 

[29]



0s

Show code

▼

...

Clean and preview the resulting data sheets!

- Cleaned sheet after applying rules

- Source column - cleaned column variable mapping

- Timepoint mapping

- Rows dropped

Cleaned data previews:

1290 cleaned rows are ready. Showing first 100 rows.

Warning: Total number of columns (73) exceeds max_columns (20) limiting to first (20) columns.

1 to 10 of 1290 entries

Filter





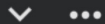
index	IRB	subid	raw_subid	Visit	Record/Subject ID:	Administered during visit number:	Initials of BSL staff:	Todays date:	1. Depression: Notes	1 a) Depressi
0	58807	s001	58807_s001	V1		Screening (Visit 1)				Severe
1	58807	s001	58807_s001	Patient Contact Log (Arm 2: Baseline/Active Study/Follow-ups/Open Label)						
2	58807	s001	58807_s001	Adverse Events/Regulatory (Arm 2: Baseline/Active Study/Follow-ups/Open Label)						
3	58807	s001	58807_s001	V2	58807_s001	Baseline (Visit 2)	NB	2021-06-23 12:04	Feels depression almost every day , had one good day on	Moderate

> 6. Download cleaned files ▾

[16]



Show code



Download multi-sheet Excel!

Please enter the file name to save this run.

Default is 58807SchatzbergRapid-HAMD_DATA_LABELS_2026-04-28_2108_cleaned.xlsx

MY-DATA

Customize exported file name

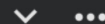
> 6. Download cleaned files ▾

[15]

✓ 34s



Show code



Please enter the file name to save this run.

Default is 58807SchatzbergRapid-HAMD_DATA_LABELS_2026-04-28_2108_cleaned.xlsx
MY-DATA.xlsx

Downloading MY-DATA.xlsx